Arista Access Node BTM-ACC-19 Routing Protocol Flap Resolution SOP

Node: BTM-ACC-19 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_9121

Description

This Standard Operating Procedure outlines the steps to troubleshoot and resolve a Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding) failure on Arista Access Node BTM-ACC-19. The Access Node is a critical component of the BTM Layout Aggregation site, responsible for forwarding traffic between different network segments. The Routing Protocol Flap is a complex phenomenon that arises due to the dynamic nature of routing protocols such as BGP, OSPF, and ISIS. These protocols continuously exchange routing information with neighboring nodes, which can lead to a high frequency of route updates, flapping, and eventual loss of OSPF adjacency or ISIS LSP flooding. The failure of the Access Node to maintain a stable routing table can cause network instability, packet loss, and increased latency. This SOP aims to identify the root cause of the Routing Protocol Flap and provide a step-by-step guide to resolve the issue, ensuring the Access Node returns to a stable state and network services are restored.

Pre-requisite Checks

- Verify that the Access Node BTM-ACC-19 is reachable and accessible remotely.

- Confirm that the site location BTM Layout Aggregation has a stable power supply and environmental conditions are within normal limits.

- Check the status of redundant links connected to the Access Node and ensure they are functioning correctly.

Potential Root Causes

Cause	Probability
BGP route churn due to incorrect routing policy configuration.	High
OSPF adjacency loss due to network topology changes or hardware issues.	Medium
ISIS LSP flooding due to misconfigured IS-IS parameters or hardware malfunctions.	Low

Diagnostic Commands & Expected Outputs

```
Command: show bgp summary, Expected Output: BGP table version 1, neighbor version 1 Neighbor V AS MsgRcvd MsgSent TblVer InQ OutQ Up/Down State | 10.10.10.1 4 65001 10 5 1 0 0 00:01:15 ESTABLISHED , Description: This command displays a summary of BGP neighbor connections, including the number of messages received and sent, and the table version. Analysis of the output can help identify any issues with BGP route churn.
```

```
Command: show ospf neighbor, Expected Output: OSPF Neighbor Status
Neighbor ID Pri State Dead Time Address Interface Vlan | 10.10.10.1 1
FULL/ - 00:01:13 10.10.10.2 eth1/1 1 , Description: This command
displays a summary of OSPF neighbor connections, including the
neighbor ID, priority, and state. Analysis of the output can help
identify any issues with OSPF adjacency loss.

Command: show isis database, Expected Output: ISIS LSP Database LSP ID
Sequence Checksum Time Age Flags | 10.10.10.1.00-00 0x00000001
0x12345678 00:01:15 00:00:01 0x2 , Description: This command displays
the ISIS LSP database, including the LSP ID, sequence number, and
checksum. Analysis of the output can help identify any issues with
ISIS LSP flooding.
```

Resolution Steps

Step 1: Step: Verify BGP routing policy configuration, Command: show bgp policy, Description: Verify that the BGP routing policy configuration is correct and does not contain any incorrect or conflicting policies that could lead to route churn.

Step 2: Step: Check OSPF network topology, Command: show ospf topology, Description: Verify that the OSPF network topology is correct and does not contain any loops or redundant links that could lead to adjacency loss.

Step 3: Step: Check IS-IS parameter configuration, Command: show isis parameters, Description: Verify that the IS-IS parameter configuration is correct and does not contain any misconfigured parameters that could lead to LSP flooding.

Post-Check Verification

```
Command: show bgp summary, Expected Output: BGP table version 1,
neighbor version 1 Neighbor V AS MsgRcvd MsgSent TblVer InQ OutQ
Up/Down State | 10.10.10.1 4 65001 10 5 1 0 0 00:01:15 ESTABLISHED ,
Description: Verify that the BGP neighbor connections are stable and
the table version has not changed.

Command: show ospf neighbor, Expected Output: OSPF Neighbor Status
Neighbor ID Pri State Dead Time Address Interface Vlan | 10.10.10.1 1
FULL/ - 00:01:13 10.10.10.2 eth1/1 1 , Description: Verify that the
OSPF neighbor connections are stable and the neighbor ID and priority
have not changed.

Command: show isis database, Expected Output: ISIS LSP Database LSP ID
Sequence Checksum Time Age Flags | 10.10.10.1.00-00 0x00000001
0x12345678 00:01:15 00:00:01 0x2 , Description: Verify that the ISIS
LSP database is stable and the LSP ID and sequence number have not
changed.
```

Roll-back Procedures

- Step: Restore original BGP routing policy configuration, Command: copy running-config startup-config, Description: Restore the original BGP routing policy configuration from the startup-config file.

- Step: Restore original OSPF network topology, Command: copy running-config startup-config, Description: Restore the original OSPF network topology from the startup-config file.

- Step: Restore original IS-IS parameter configuration, Command: copy running-config startup-config, Description: Restore the original IS-IS parameter configuration from the startup-config file.